

Identity at the Center: How AI and Cloud Adoption Are Reshaping Enterprise Attack Surfaces

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Executive Summary

Identity has become a central component of modern enterprise security as organizations increasingly rely on cloud platforms, software-as-a-service (SaaS) applications, and AI-enabled technologies to support business operations. Centralized identity providers, federated authentication systems, and single sign-on solutions now govern access to a growing number of enterprise resources, increasing both the value of trusted identities and the potential impact of identity compromise.

At the same time, generative AI is enhancing many of the social engineering and deception techniques historically used by threat actors to obtain access. Large language models, voice synthesis technologies, and deepfake generation tools enable adversaries to create more convincing phishing campaigns, impersonate trusted individuals, and conduct social engineering operations at greater scale. While these capabilities do not fundamentally change attacker objectives, they may increase the effectiveness of identity-focused attacks by improving deception and reducing operational barriers.

The rapid adoption of AI assistants, retrieval-augmented generation (RAG) applications, cloud-native AI services, and autonomous agents further increases enterprise reliance on identity systems. Many AI-enabled workflows depend on identities, permissions, API credentials, and service accounts to access enterprise data and business functions. Consequently, the compromise of a trusted identity may provide adversaries with access not only to traditional enterprise resources, but also to emerging AI-enabled systems and workflows.

This report assesses the evolving relationship between identity security, cloud adoption, and artificial intelligence. Based on the trends examined throughout this report, we assess that identity security is emerging as a foundational component of AI security. As organizations continue integrating AI technologies into business operations, the effectiveness of identity-focused attacks and the strategic importance of identity governance are likely to continue increasing.

Key Takeaways

- Identity-focused attacks remain one of the most effective methods for obtaining enterprise access.
- Generative AI is amplifying social engineering, phishing, impersonation, and other identity-focused attack techniques.
- Cloud and AI adoption are increasing organizational reliance on centralized identity systems and machine identities.
- Identity governance, access management, and identity-centric detection capabilities are becoming increasingly important components of enterprise AI security.

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Introduction

Identity has long served as the foundation of enterprise security, governing access to users, applications, cloud resources, and sensitive data. Over the past decade, threat actors have consistently demonstrated that compromising a trusted identity often provides a more efficient path to enterprise access than exploiting technical vulnerabilities. At the same time, organizations are increasingly adopting cloud-native services, AI-powered applications, and autonomous agents that rely heavily on identity systems for authentication and authorization. As a result, digital identity has evolved from a security control into a critical enterprise dependency, making it an increasingly attractive target for adversaries.

The rapid adoption of generative AI technologies is further reshaping the identity threat landscape. Large language models, voice synthesis platforms, and deepfake generation tools have lowered the barriers to conducting sophisticated social engineering operations, enabling attackers to create highly convincing phishing campaigns, impersonate trusted individuals, and automate reconnaissance at scale. While these capabilities do not fundamentally alter the objectives of threat actors, they may significantly enhance an adversary's ability to obtain valid credentials, manipulate identity verification processes, and exploit trust-based security controls.

Simultaneously, enterprises are increasingly integrating AI capabilities into business processes, cloud platforms, and productivity workflows. AI assistants, copilots, retrieval-augmented generation (RAG) systems, and autonomous agents increasingly operate with access to sensitive enterprise data and business functions. In many environments, identity systems serve as the primary mechanism controlling access to these resources. Consequently, successful identity compromise may provide attackers with access not only to traditional enterprise assets, but also to emerging AI-enabled systems and workflows.

This report assesses how generative AI is amplifying identity-focused attacks while increasing enterprise reliance on identity as a foundational security control. It examines emerging attack techniques, evolving threat actor tradecraft, and the growing convergence of identity, cloud, and AI ecosystems. The report further evaluates the implications of these trends for organizations seeking to secure cloud-native and AI-enabled environments.

1.0 The Evolution of Identity-Focused Attacks

Identity-focused attack techniques have become an increasingly common method of initial access for many threat actors due to their effectiveness and scalability. Rather than relying exclusively on software vulnerabilities or custom malware, adversaries increasingly target the people, processes, and identity systems that govern access to enterprise environments. Recent intrusions involving organizations such as MGM Resorts and Caesars Entertainment demonstrated how social engineering, help desk manipulation, and identity abuse can provide attackers with access to critical enterprise resources. These incidents highlight a broader shift in the threat landscape, where compromising a trusted identity often provides a more efficient path to enterprise access than exploiting technical vulnerabilities.

Historically, identity-focused attacks primarily centered on credential theft through phishing campaigns, password spraying, and business email compromise (BEC). While these techniques remain prevalent, the expansion of cloud computing, software-as-a-service (SaaS) applications, and centralized identity providers has fundamentally changed the value of a compromised identity. Modern enterprise users often access dozens of business-critical applications through a single authentication workflow, allowing adversaries to leverage one compromised account to gain access to email, collaboration platforms, cloud resources, and sensitive corporate data. As organizations increasingly consolidate access through identity providers such as Okta and Microsoft Entra ID, identity systems have become attractive targets for threat actors seeking broad and scalable access to enterprise environments.

Recent campaigns have demonstrated the continued effectiveness of identity-focused attack techniques as a means of obtaining enterprise access. High-profile incidents, including the 2023 intrusions affecting MGM Resorts and Caesars Entertainment, highlighted how threat actors could leverage social engineering, help desk manipulation, and identity abuse to gain access to enterprise environments without exploiting software vulnerabilities.¹²³ These incidents reflected a broader shift in attacker tradecraft, where the compromise of a trusted identity often provides a more efficient path to enterprise access than traditional exploitation methods. As enterprise environments have become increasingly dependent on centralized identity providers, attackers have shifted their focus toward obtaining legitimate credentials, authenticated sessions, and trusted access rather than relying solely on malware or software exploits.⁴ Recent reporting from identity providers and industry organizations indicates that adversaries continue to abuse social engineering, credential theft, MFA manipulation, session hijacking, and OAuth-based attacks to

gain access to cloud and SaaS environments.⁴⁵ As organizations consolidate authentication through platforms such as Okta and Microsoft Entra ID, the compromise of a single trusted identity may provide access to numerous business-critical systems, making identity infrastructure an increasingly valuable target for financially motivated threat actors and other adversaries.⁴⁵

Identity-focused attacks have continued to evolve beyond traditional phishing and credential theft. Modern adversaries increasingly target the trust mechanisms underpinning enterprise identity systems through techniques such as MFA reset abuse, OAuth consent phishing, session hijacking, adversary-in-the-middle (AiTM) attacks, and token theft. Rather than defeating authentication controls directly, these techniques seek to obtain legitimate access by exploiting user trust, administrative processes, or authenticated sessions. According to Okta, attackers increasingly focus on acquiring valid credentials, session tokens, and privileged access through social engineering and authentication abuse rather than exploiting software vulnerabilities.⁴ OAuth-based attacks have proven particularly effective because they allow adversaries to obtain persistent access through malicious application consent without necessarily compromising a user's password.⁶ Similarly, session hijacking and token theft enable attackers to bypass traditional authentication controls by abusing already authenticated sessions and trusted identity relationships.⁷ As organizations continue to adopt cloud-native applications, federated identity architectures, and single sign-on solutions, these techniques provide threat actors with scalable methods for accessing enterprise resources while blending into legitimate user activity.⁴⁵

Section 2

How Generative AI Is Enhancing Identity Attacks

While identity-focused attacks are not new, generative AI has the potential to significantly enhance their effectiveness, scale, and accessibility. Large language models, voice synthesis platforms, and deepfake generation tools enable threat actors to automate portions of the attack lifecycle, create more convincing social engineering content, and impersonate trusted individuals with increasing realism. Although these technologies do not fundamentally alter the objectives of adversaries, they may reduce barriers to entry and increase the likelihood of successful identity compromise by enhancing existing attack techniques rather than replacing them.⁸⁹¹⁰

2.1 AI-Enhanced Social Engineering and Phishing

Social engineering remains one of the most commonly observed and effective methods for obtaining enterprise access because it targets human behavior rather than technical vulnerabilities. Generative AI is increasing the effectiveness of these attacks by enabling threat actors to create more convincing, personalized, and scalable phishing campaigns. Large

language models can rapidly generate realistic emails, text messages, and other communications that mimic professional writing styles, incorporate organization-specific details, and adapt messaging to individual targets.⁸⁹¹⁰

In addition to content generation, AI assists adversaries in collecting and analyzing publicly available information about potential victims. By rapidly processing large volumes of data from social media, professional networking sites, and other online sources, AI tools can help attackers identify relationships, communication patterns, and contextual details that can be leveraged during social engineering operations.⁸⁹ This intelligence can then be used to craft highly targeted phishing lures that appear legitimate and relevant to the intended recipient.

Historically, attackers often faced a tradeoff between scale and personalization. Broad phishing campaigns could target large numbers of victims but frequently lacked credibility, while highly targeted attacks required significant time and effort to develop. Generative AI increasingly reduces this tradeoff by enabling adversaries to produce personalized content at scale. According to multiple security researchers, AI-generated phishing content can exhibit improved grammar, natural language fluency, and localization capabilities, making malicious communications more difficult for users to identify.⁸⁹¹⁰¹¹¹² As a result, generative AI may serve as a force multiplier for social engineering operations, increasing both the volume and effectiveness of identity-focused attacks while lowering the resources required to conduct them.

2.2 Deepfakes and Voice Cloning

While generative AI enhances traditional phishing and social engineering operations, advances in deepfake and voice synthesis technologies present a distinct challenge to identity verification and trust-based security processes. Deepfakes leverage artificial intelligence to generate realistic audio, video, and visual content capable of impersonating trusted individuals with increasing accuracy. Unlike conventional social engineering attacks, which primarily rely on written communication, deepfakes enable adversaries to exploit voice and visual cues that organizations often use to establish trust and verify identity.¹³¹⁴

The barriers to creating convincing impersonations have decreased significantly in recent years. Modern voice cloning tools can replicate an individual's speech patterns, tone, and cadence using only a small sample of audio, while deepfake technologies can generate realistic video content capable of mimicking facial expressions and body language. According to multiple security researchers, these capabilities have already been leveraged in fraud schemes, business email compromise (BEC) operations, and other forms of social engineering designed to manipulate employees into transferring funds, disclosing sensitive information, or granting access to enterprise resources.¹³¹⁴¹⁵

The implications for enterprise security extend beyond financial fraud. Many organizations continue to rely on voice-based verification procedures, executive approval processes, help desk interactions, and other trust-based mechanisms as part of identity and access management workflows. As deepfake and voice cloning technologies become more accessible, adversaries may increasingly exploit these processes to facilitate account takeover, bypass

identity verification controls, and support broader intrusion activity. The Financial Services Sector Coordinating Council (FSSCC) has identified AI-enabled impersonation and synthetic identity abuse as emerging risks capable of undermining traditional authentication and identity assurance mechanisms.⁵ Consequently, organizations may need to reevaluate identity verification procedures that rely heavily on voice, video, or other forms of human recognition as indicators of trust.

2.3 AI Brands as Bait

Beyond leveraging generative AI to improve social engineering operations, threat actors have increasingly begun exploiting trust in AI technologies themselves. As enterprise and consumer adoption of AI platforms continues to grow, popular services such as ChatGPT, Microsoft Copilot, Claude, Gemini, and DeepSeek have become recognizable and trusted brands. According to Microsoft Threat Intelligence, adversaries are increasingly using AI-related themes, branding, and lures to facilitate phishing campaigns, malware distribution, and credential theft operations.¹⁶

These campaigns frequently capitalize on user curiosity surrounding emerging AI technologies by impersonating legitimate AI providers, advertising fake AI tools, or directing victims to malicious websites masquerading as trusted AI platforms. In many cases, the objective is not to compromise the AI platform itself, but rather to exploit the trust users place in these services to facilitate identity compromise and malware delivery. This trend reflects a broader evolution in social engineering tradecraft, where attackers increasingly align lures with current technological trends and user interests to improve the likelihood of success.

The growing adoption of enterprise AI solutions may further expand these opportunities. As organizations integrate AI assistants, copilots, and AI-enabled workflows into daily operations, employees may increasingly rely on AI systems for information retrieval, research, productivity tasks, and decision support. This growing reliance creates additional opportunities for threat actors to exploit trust in AI platforms and services as part of broader identity-focused attack campaigns. Beyond traditional brand impersonation, adversaries may also seek to manipulate the information ecosystems that influence AI-generated outputs and recommendations, increasing the likelihood that users interact with malicious content, fraudulent services, or phishing infrastructure presented through trusted AI interfaces.¹⁶¹⁹

Emerging reporting suggests that threat actors may also seek to manipulate the information sources consumed by AI systems themselves. Researchers and industry reporting have highlighted scenarios in which fraudulent websites, scam storefronts, or attacker-controlled content are optimized to appear within AI-generated recommendations and search results. Rather than directly targeting the user through traditional phishing channels, these techniques seek to exploit trust in the AI system's recommendations, potentially increasing the credibility of malicious links and fraudulent services. This evolution suggests that attackers may increasingly target not only AI users, but also the broader information ecosystems that influence AI-generated outputs and recommendations.¹⁹

2.4 Why This Matters for Identity

The significance of these developments extends beyond the emergence of new attack techniques. Many modern intrusion campaigns already rely on social engineering, credential theft, MFA abuse, and other identity-focused tactics as primary means of obtaining enterprise access. The capabilities discussed throughout this section—including AI-enhanced phishing, personalized social engineering, deepfake impersonation, voice cloning, and AI-themed lures—primarily serve to increase the effectiveness of existing identity attack techniques rather than introduce entirely new methods of compromise.⁸⁹¹⁰¹³¹⁴¹⁶

As organizations increasingly depend on centralized identity providers, cloud-based authentication systems, and federated access architectures, successful identity compromise can provide access to a growing number of enterprise resources. Generative AI may therefore function as a force multiplier for identity-focused attacks by improving deception, reducing operational costs, and enabling greater levels of scale and personalization. Consequently, organizations should increasingly view identity security as a foundational component of AI security, as the compromise of trusted identities may provide adversaries with access not only to users and applications, but also to cloud services, enterprise data, and emerging AI-enabled systems.

Section 3

Identity as the Control Plane for Cloud and AI

The growing effectiveness of identity-focused attacks is occurring alongside a fundamental shift in enterprise architecture. As organizations migrate workloads to the cloud and adopt AI-enabled applications, identity systems increasingly serve as the primary mechanism governing access to users, services, data, and business-critical resources. Rather than functioning solely as authentication tools, modern identity platforms have evolved into centralized control planes that determine who—and increasingly what—can access enterprise systems. Consequently, the security of identity infrastructure has become increasingly intertwined with the security of cloud and AI environments.

3.1 Identity as the Foundation of Cloud Access

The adoption of cloud computing has significantly increased the value and strategic importance of enterprise identities. Historically, user accounts were often limited to a relatively small number of on-premises systems and applications. However, the widespread adoption of cloud platforms,

software-as-a-service (SaaS) applications, and remote work technologies has fundamentally changed how organizations manage access. Modern enterprises increasingly rely on centralized identity providers and single sign-on (SSO) solutions to authenticate users across numerous cloud services, collaboration platforms, business applications, and data repositories. As a result, identity has evolved beyond a mechanism for user authentication and now serves as the primary control layer governing access to enterprise resources.⁴⁵¹⁷

This shift has created both operational efficiencies and security challenges. A single trusted identity may now provide access to email platforms, file storage services, cloud infrastructure, productivity applications, customer data, and other business-critical systems through a unified authentication workflow. While this model simplifies access management for organizations and users, it also concentrates risk. The compromise of a single identity may provide adversaries with access to numerous downstream resources without requiring additional exploitation activity. Consequently, identity systems have become high-value targets for threat actors seeking scalable and persistent access to enterprise environments.⁴⁵¹⁷

3.2 The Emergence of AI Identities and Agents

The rapid adoption of enterprise AI introduces identity challenges that extend beyond traditional user accounts. Historically, identity and access management programs primarily focused on governing human users, including employees, contractors, and third-party partners. However, modern cloud and AI environments increasingly rely on machine identities, service accounts, API credentials, and autonomous systems that require authenticated access to enterprise resources. As organizations deploy AI assistants, copilots, retrieval-augmented generation (RAG) systems, and agentic workflows, identity systems must increasingly govern not only who can access enterprise resources, but also what can access them.⁵¹⁷¹⁸

Many AI-enabled applications operate by accessing data, services, and business processes on behalf of users. For example, enterprise AI platforms may retrieve information from cloud storage repositories, collaboration platforms, customer relationship management systems, internal knowledge bases, or business applications. To perform these functions, AI systems often rely on delegated permissions, service accounts, API keys, OAuth tokens, or other forms of machine authentication. Consequently, AI systems increasingly possess identities and privileges that must be managed, monitored, and secured in the same manner as traditional user accounts.⁵¹⁷

This shift has important security implications. The compromise of an AI system, service account, or machine identity may provide adversaries with access to sensitive data, automated workflows, or business functions without directly compromising a human user account. As AI adoption accelerates, organizations may need to expand traditional identity security programs to address the growing ecosystem of non-human identities operating throughout cloud and AI environments.¹⁷¹⁸

3.3 Identity as an AI Security Control

Discussions surrounding AI security frequently focus on model vulnerabilities, prompt injection, adversarial manipulation, and other attacks directed at AI systems themselves. While these risks are important, they may overlook a more fundamental reality: enterprise AI systems ultimately depend on identity and access controls to govern who can interact with models, access underlying data, invoke automated actions, and manage AI-enabled workflows. Consequently, the compromise of a trusted identity may provide adversaries with indirect access to AI resources without requiring direct attacks against the models themselves.⁵¹⁷¹⁸

In many enterprise environments, AI systems are integrated with cloud services, collaboration platforms, knowledge repositories, and business applications through identity-based access mechanisms. As a result, the permissions assigned to users, service accounts, and AI agents often determine the scope of information and functionality available to AI-enabled workflows. A compromised identity may therefore enable adversaries to access sensitive datasets, manipulate AI-generated outputs, retrieve proprietary information, or abuse automated processes operating on behalf of trusted users. This dynamic suggests that identity compromise may represent a significant pathway to AI-related risk, particularly as organizations expand the deployment of AI assistants, copilots, agentic systems, and retrieval-augmented generation applications.

As enterprises continue integrating AI capabilities into critical business functions, identity security may increasingly serve as a foundational AI security control. Organizations that focus exclusively on securing AI models while neglecting identity governance, access management, and authentication controls may overlook one of the most practical paths through which adversaries can gain access to AI-enabled systems and the data they depend upon. Based on the trends discussed throughout this report, we assess that the convergence of identity, cloud, and AI ecosystems will continue to increase the strategic importance of identity security as a component of enterprise AI risk management.⁵¹⁷¹⁸

4. Detection and Mitigation Considerations

The convergence of identity, cloud, and AI technologies presents both new opportunities and new security challenges for organizations. While many of the attack techniques discussed throughout this report are not entirely new, generative AI has the potential to increase their scale, effectiveness, and accessibility. At the same time, the growing reliance on identity systems to govern access to cloud resources, enterprise data, and AI-enabled workflows increases the potential impact of successful identity compromise. Consequently, organizations should consider adapting detection, monitoring, and identity governance strategies to address the evolving relationship between identity security and AI risk.

4.1 Detecting Identity-Focused Attacks

Detecting identity-focused attacks presents unique challenges for defenders because these operations frequently rely on legitimate credentials, authenticated sessions, and trusted access pathways rather than malware or software exploitation. Unlike traditional intrusion activity, where malicious code or suspicious network traffic may provide clear indicators of compromise, identity-based attacks often blend into normal user behavior. Threat actors increasingly seek to obtain valid credentials, session tokens, and trusted access through phishing, social engineering, OAuth abuse, MFA manipulation, and other identity-focused techniques. Once access has been obtained, adversaries may interact with enterprise systems in ways that appear indistinguishable from legitimate users, reducing the effectiveness of traditional endpoint-centric detection strategies.⁴⁵

As a result, organizations should increasingly prioritize identity telemetry as a primary source of detection and investigation. Security teams should monitor for behaviors commonly associated with identity abuse, including unusual authentication activity, suspicious MFA reset requests, anomalous account recovery events, unauthorized OAuth consent grants, session hijacking attempts, impossible travel events, and unexpected privilege escalation. Particular attention should be given to newly enrolled authentication devices, unfamiliar application consent activity, and help desk interactions involving identity recovery workflows, which have repeatedly been targeted by threat actors seeking initial access through social engineering and trust-based processes.¹²³⁴

The growing adoption of cloud services further increases the importance of identity visibility. Because centralized identity providers often serve as gateways to numerous cloud platforms, SaaS applications, and business-critical resources, suspicious authentication activity should be evaluated within the broader context of downstream resource access. A successful identity compromise may enable access to email systems, cloud storage platforms, collaboration tools, enterprise applications, and AI-enabled workflows through a single authentication event. Consequently, organizations should consider correlating identity events with cloud activity, privilege usage, and data access patterns to identify potentially malicious behavior operating under the guise of legitimate access.⁴⁵¹⁷

The emergence of AI-enhanced social engineering may further complicate detection efforts. Generative AI enables threat actors to produce highly convincing phishing campaigns, impersonation attempts, and other forms of identity-focused deception at scale. As these techniques continue to improve, organizations may increasingly face adversaries capable of obtaining valid access without relying on traditional malware or exploit-based tradecraft. We assess that identity telemetry, behavioral analytics, and continuous monitoring of authentication activity will become increasingly important components of enterprise detection strategies as the convergence of identity, cloud, and AI ecosystems continues to reshape the threat landscape.⁴⁵⁸⁹¹⁰

4.2 Securing AI Platforms and AI Identities

As organizations integrate AI capabilities into enterprise environments, security teams must increasingly consider how AI systems access data, services, and business processes. Unlike traditional applications, many AI-enabled systems operate across multiple platforms, repositories, and workflows while relying on a combination of user identities, service accounts, API credentials, and machine identities. Consequently, securing AI environments requires visibility not only into the AI models themselves, but also into the identities and permissions that enable AI systems to function.⁵¹⁷¹⁸

Organizations deploying retrieval-augmented generation (RAG) applications should consider monitoring for unusual document retrieval activity, excessive query volumes, unauthorized access to knowledge repositories, and unexpected interactions with sensitive data sources. Because RAG systems often retrieve information from cloud storage platforms, collaboration tools, and internal knowledge bases, the permissions granted to these systems may directly influence the data exposed to users. Similarly, cloud-native AI platforms such as Amazon Bedrock rely heavily on identity and access management controls to govern access to models, APIs, datasets, and integrated cloud services. Security teams should monitor IAM role assignments, API usage, model invocation activity, service account behavior, and privilege changes that could indicate misuse or unauthorized access.

Broader AI ecosystems such as Hugging Face introduce additional considerations related to API tokens, model repositories, third-party integrations, and collaborative development workflows. Organizations should maintain visibility into repository access, exposed credentials, unauthorized token usage, and external integrations capable of introducing risk into AI development environments. While these platforms differ in functionality and deployment models, they share a common dependency on identity and access controls to regulate interactions between users, applications, data, and AI services.

We assess that identity governance will increasingly serve as a foundational component of AI security. As enterprises continue adopting AI assistants, agentic workflows, RAG applications, cloud-based AI services, and autonomous agents, the permissions assigned to users, service accounts, and machine identities may ultimately determine the scope of data and functionality available to AI systems. Consequently, organizations should view identity management, credential protection, access monitoring, and least-privilege enforcement as critical components of securing modern AI environments.⁵¹⁷¹⁸

4.3 Preparing for AI-Enhanced Social Engineering

The growing use of generative AI, voice synthesis, and deepfake technologies may require organizations to reevaluate traditional trust-based verification processes. Many business workflows continue to rely on verbal approvals, executive requests, help desk interactions, and other forms of identity verification that assume voice, appearance, or familiarity are reliable indicators of authenticity. As AI-generated impersonation capabilities continue to improve, organizations should consider implementing layered verification procedures for sensitive actions, including out-of-band validation for financial transactions, account recovery requests,

privileged access changes, and other high-risk activities. Security teams may also benefit from incorporating deepfake awareness into security training programs and developing executive impersonation response procedures that account for AI-enabled deception. While no single control is likely to eliminate the risk posed by AI-enhanced social engineering, strong multi-factor authentication, conditional access policies, identity verification safeguards, and well-defined escalation procedures may help reduce opportunities for adversaries to exploit trust-based processes. We assess that organizations should increasingly assume that voice, video, and written communications can be convincingly impersonated and adapt identity verification practices accordingly.⁵¹³¹⁴¹⁵

Table 1. Defensive Considerations for Identity, Cloud, and AI Environments

Focus Area	Risks Discussed in This Report	Defensive Considerations
Identity-Focused Attacks	Credential theft, MFA abuse, OAuth abuse, session hijacking, help desk manipulation	Monitor authentication activity, MFA resets, OAuth consent grants, account recovery events, privilege escalation, and identity telemetry
Cloud Identity Infrastructure	Centralized access through SSO, federated identity, cloud authentication systems	Correlate identity events with cloud activity, monitor downstream resource access, review privileged account activity, enforce least privilege
AI Platforms and Services	RAG applications, Amazon Bedrock, AI assistants, machine identities	Monitor IAM roles, API usage, model invocation activity, service account permissions, document retrieval activity, and access to sensitive data sources

AI Development Ecosystems	Hugging Face repositories, API tokens, third-party integrations, exposed credentials	Monitor repository access, token usage, credential exposure, external integrations, and development environment permissions
AI-Enhanced Social Engineering	AI-generated phishing, deepfakes, voice cloning, executive impersonation	Implement layered verification procedures, out-of-band validation, strong MFA, conditional access policies, and deepfake awareness training
AI Identities and Agents	Service accounts, machine identities, autonomous agents, delegated permissions	Enforce identity governance, credential management, access monitoring, and least-privilege controls for non-human identities

The defensive considerations outlined above are intended to highlight areas organizations may prioritize as identity, cloud, and AI ecosystems continue to converge. While individual implementation approaches will vary by organization, visibility into identity activity and access controls remains a recurring theme across each risk area examined in this report.

Key Findings

Finding 1: Identity Remains One of the Most Effective Intrusion Vectors

Identity-focused attacks continue to provide adversaries with a highly effective means of obtaining enterprise access. Rather than relying exclusively on software vulnerabilities or malware, threat actors increasingly leverage social engineering, credential theft, OAuth abuse, MFA manipulation, and other identity-focused techniques to obtain legitimate access. The continued success of these methods demonstrates that compromising a trusted identity often remains a more efficient path to enterprise access than exploiting technical vulnerabilities.

Finding 2: Generative AI Is Amplifying Social Engineering and Identity-Focused Attacks

Generative AI is acting as a force multiplier for social engineering operations by improving the realism, scalability, and personalization of phishing campaigns, impersonation attempts, and other forms of identity-focused deception. While AI does not fundamentally change attacker objectives, it may significantly reduce the resources required to conduct convincing social engineering operations at scale. Deepfake technologies, voice cloning capabilities, and AI-enhanced phishing techniques further increase the potential for successful identity compromise.

Finding 3: Cloud and AI Adoption Are Increasing Enterprise Reliance on Identity Systems

The continued adoption of cloud platforms, SaaS applications, AI assistants, and retrieval-augmented generation systems has elevated the strategic importance of enterprise identity infrastructure. Modern organizations increasingly rely on centralized identity providers, service accounts, API credentials, and machine identities to govern access to critical resources. As a result, the compromise of a trusted identity may provide access to a growing number of cloud services, enterprise datasets, and AI-enabled workflows.

Finding 4: Identity Security Is Emerging as a Foundational AI Security Control

Although discussions surrounding AI security frequently focus on model vulnerabilities and adversarial attacks, enterprise AI systems ultimately depend on identity and access controls to govern access to data, services, and automated actions. As organizations continue integrating AI capabilities into business processes, the permissions assigned to users, service accounts, AI agents, and machine identities may increasingly determine the scope of AI-related risk. Consequently, identity governance, credential protection, and access management should be viewed as foundational components of enterprise AI security strategies.

Conclusion

Identity has long represented a critical component of enterprise security, but the convergence of cloud computing, artificial intelligence, and modern identity architectures has significantly increased its strategic importance. As organizations continue centralizing access through identity providers, federated authentication systems, and cloud-based services, trusted identities increasingly serve as gateways to enterprise applications, sensitive data, business processes, and AI-enabled workflows. Consequently, the compromise of a trusted identity may provide adversaries with access to a growing number of interconnected resources through a single point of entry.

At the same time, generative AI is enhancing many of the social engineering and deception techniques that threat actors have historically used to obtain access. AI-generated phishing campaigns, voice cloning technologies, deepfake impersonation, and AI-themed lures may increase the scale, realism, and effectiveness of identity-focused attacks while lowering the resources required to conduct them. Although these technologies do not fundamentally alter attacker objectives, they may significantly improve an adversary's ability to exploit trust-based security controls and identity verification processes.

The growing adoption of AI assistants, retrieval-augmented generation systems, cloud-native AI services, and autonomous agents further reinforces the importance of identity security. These technologies increasingly depend on identity and access management systems to govern access to data, services, and automated actions. As a result, identity compromise may provide a practical pathway to AI-related risk without requiring direct attacks against AI models themselves. Based on the trends examined throughout this report, we assess that the convergence of identity, cloud, and AI ecosystems will continue to elevate identity security as a foundational component of enterprise security and AI risk management. Organizations that prioritize identity governance, access management, and identity-centric detection capabilities will likely be better positioned to address the evolving threat landscape.

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